

Akash Bachhar

Robotics Engineer

I'm **Akash Bachhar**, an engineer driven by one question: what does it take to build machines that can perceive, reason, move, and collaborate alongside humans?

With a background in [Mechanical Engineering from NIT Durgapur](#), I've built my journey around the intersection of **robotics, mechatronics, and computer vision-based learning**. My fascination with humanoid robotics began in my early teenage years and has since grown into a broader interest in intelligent machines and the technologies that bring them to life. Today, that curiosity fuels both my professional work and personal projects, where I combine mechanics, electronics, software, and AI to turn ideas into working systems.

I previously worked as a **Visiting Researcher** at [Central Mechanical Engineering Research Institute \(CMERI\)](#) and as a **Summer Intern** at [R&DE, DRDO](#), and as a **Remote Research Collaborator** at [Aalborg University, Department of Electronic Systems](#).

Outside of my core research, I explore adjacent areas including **full-stack development, augmented reality, and game development** — building experimental tools and prototypes. I'm also an avid reader and enjoy photography and travel.

[Email](#)[GitHub](#)[Google Scholar](#)[X](#)[LinkedIn](#)

Research Vision



Vision-Guided Robot Learning

Combining perception and machine learning to help robots acquire complex skills from data and experience.



Dexterous Manipulation

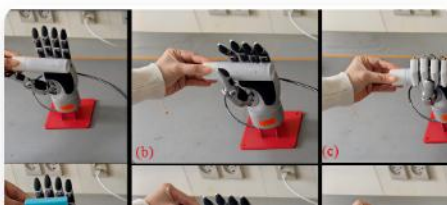
Designing control strategies for precise, adaptive, human-like grasping in prosthetic systems.



Human-Robot Collaboration

Building remote manipulation systems, exoskeletons, and human-in-the-loop robotic control.

Publications



[Per-Finger Prosthetic Grasp Planning Using Object-Aligned Bounding Box Representation and VLM-Driven Object Selection](#)

Shifa Sulaiman, Akash Bachhar, Ming Shen, Simon Bøgh, Luigi Bibbo

Applied Sciences 16 (12), 5736

A Vision-Language Model identifies the target object and triggers grasping only on confident recognition, enabling intent aware control. An object-aligned bounding box from the



recognition, enabling intent-aware control. An object-aligned bounding box from the segmented point cloud supports accurate distance and collision queries, with per-finger contact poses planned independently and solved via Damped Least Squares inverse kinematics. Validated on the physical Linker Hand O7, achieving 93.2% segmentation accuracy (93.4% in simulation) with stable, real-time grasp execution.

 Paper



Hierarchical Vision-Based and Sequential Deep Learning Architectures for Subject-Independent Locomotion Mode Detection, Speed, and Gait Phase Estimation

Debadrata Sarkar, Akash Bachhar, Aman Arora

Manuscript in Preparation



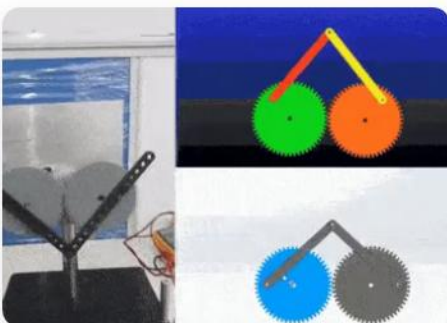
Vision-Guided Grasp Planning for Prosthetic Hands with AABB-Based Object Representation

Shifa Sulaiman, Akash Bachhar, Ming Shen, Simon Bøgh

Robotics 15 (1), 22

Introduces a vision-guided grasp planning framework for prosthetic hands in unstructured environments. A wrist-mounted camera and Bounding Volume Hierarchy (BVH) algorithm segment the target object and compute its bounding box, with grasp trajectories generated via RRT* and fingertip poses selected by minimum Euclidean distance to the object point cloud. Fingers are planned independently using Damped Least Squares inverse kinematics for adaptive, object-specific grasps. Validated via sim-to-real transfer on the physical Linker Hand O7.

 Paper



Single-Actuator Driven Symmetric Five-Bar Mechanism for Producing Reciprocal Rectilinear High-Deflection Motion for Pumping Bag-Valve-Mask (BVM)

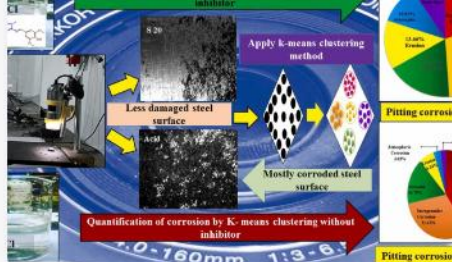
Nilanjan Chattaraj, Akash Bachhar

Journal of Vibration Engineering & Technologies (Accepted)

This work presents a novel single-actuator driven symmetric five-bar mechanism capable of generating high-deflection reciprocal rectilinear motion, specifically designed for automated Bag-Valve-Mask (BVM) pumping applications. The mechanism achieves precise motion control with reduced mechanical complexity, offering a compact, reliable, and cost-effective solution for emergency respiratory support systems.



Quantification and inhibition of corrosion on mild steel by the synthesized inhibitor through k-means clustering and electrochemical study: A dual approach



Surya Sarkar, Akash Bachhar, Sukdeb Mandal, Samik Dutta, Priyabrata Banerjee

International Journal of Electrochemical Science 19 (9), 100746

Developed a k-means image segmentation method for rapid, accurate quantification of corrosion damage on mild steel surfaces, and evaluated the inhibition efficiency of four synthesized inhibitors via electrochemical experiments. Experimental results closely matched theoretical predictions, validating the image processing approach as a reliable tool for corrosion inhibition assessment.

 Paper

Projects



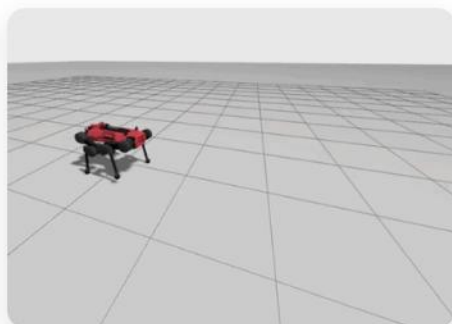
Force-Reflecting Hand Exoskeleton for Master-Slave Remote Handling (Final Year Thesis)

Developed a DRDO-funded hand exoskeleton for master-slave remote manipulation, providing bidirectional kinesthetic force feedback via IMUs and potentiometers. Master-side movements are kinematically retargeted to an Allegro robotic hand, with grasping forces mirrored back in real time. Includes a head-mounted display and 3-axis servo gimbal camera for immersive remote operation.



MARCUS - Magnetic Adhesion Remote Controlled Utility Surface-Crawler

Developed a flameproof, weather-resistant surface-crawling robot for LPG Horton sphere maintenance, using permanent magnetic adhesion optimized for curved surfaces and pneumatic drive motors. Wirelessly controlled via joystick, with a modular magnetic system ensuring grip across varying sphere geometries. Enables cleaning, painting, and NDT tasks, cutting manpower by 80% and operational time by 90% while eliminating height-related risks and scaffolding costs.

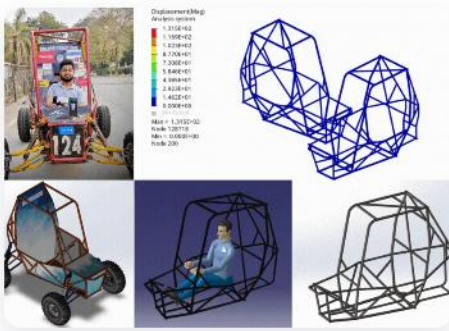


Quadruped Locomotion: Kinematics, Control, and Gait Generation in ROS 2 (Summer Internship)

Built a full locomotion stack for the ANYmal quadruped in ROS 2 and Gazebo — closed-form kinematics, joint control, and gait generation. Derived analytical forward/inverse kinematics for all four legs directly from the URDF, replacing numerical solvers with exact closed-form expressions. Designed a PD-plus-gravity-feedforward joint controller and an IMU-driven PID stabilizer, tuned against measured tracking error and sensor-frame quirks. Generated static and dynamic gaits (walk, trot) from a shared duty-factor/phase-offset model, with a diagnostic tool to quantify gait quality from live telemetry.

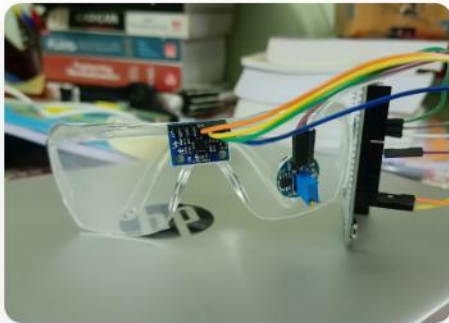
 Code

Roll Cage Design and Manufacturing for BAJA SAE 2023



Led the design and fabrication of a lightweight, high-strength roll cage as Roll Cage Head of Team NDORS, NIT Durgapur, for BAJA SAE India 2023. Optimized material and structural design per the rulebook, achieving 20% cost and 30% weight reduction over the previous year. Conducted static (impact, bump) and dynamic (rollover, collision) analyses, and validated manufacturing through weld strength and drop testing. The team placed 2nd in CAE Presentation and 3rd in All-Terrain Performance among IITs and NITs, ranking 16th and 19th nationally.

 Presentation



SpecsX – Gyroscopic Smart Glasses for Hands-Free Computer Control

Built a gyroscope-equipped eyewear system enabling specially-abled individuals to control a computer's mouse and keyboard through head movements, with clicks triggered by detecting eye blinks and their duration. SpecsX enables independent, confident digital access for education and communication.

 Code

Mini Hobby Projects



Quadruped Locomotion with PPO-Based Reinforcement Learning

 Code



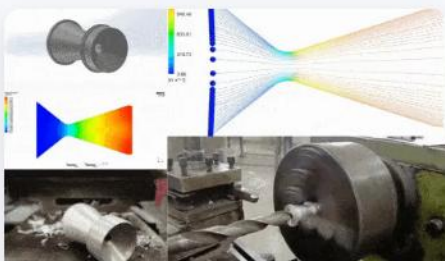
ArmX – Precision Pick-and-Place with UR5e Robot using MoveIt

 Code



Third Eye – Real-Time Driver Drowsiness Detection and Alert System

 Code



Design, Simulation, and Fabrication of High-Performance Solid Rocket Nozzle



HandX – Robotic Hand Control via Image Processing-based Hand Posture Detection



SpeedX – Immersive Gyroscope-Controlled Racing Game for Medical Rehabilitation

 Code